

Koa Shen

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Hiring Team

Foundation

Re: Mechanical Engineering Internship

Dear Foundation Hiring Team,

I am a mechanical engineering student at UC Santa Barbara applying for a mechanical engineering internship at Foundation. I am drawn to Foundation's work on autonomous machines because it brings mechanical systems, actuation, and autonomy together in robots intended to do useful work in the physical world. That is the kind of engineering I want to contribute to: translating a system-level goal into hardware that can be built, tested, and made reliable.

At SLAC National Accelerator Laboratory, I built an open-source Python/Drake digital-twin framework for tightly packaged motion assemblies. Applied to a 30-DOF, 305-body optics assembly, millisecond collision checks exposed at least five real pinch points and informed redesigns to reduce interference risk around detectors valued at more than \$100,000. This work taught me to use modeling as a practical design tool, connecting geometry and motion constraints to changes engineers can implement on real hardware.

In a personal robotics project, I designed a compact 15:1 cycloidal reducer packaged within a NEMA 17 motor's 42 mm square face, then built C++ firmware with encoder feedback for closed-loop position correction and missed-step detection. Bench testing validated the feedback and compensation on a single-joint setup. With Gaucho Racing, I have also CNC-machined 33 suspension components, holding critical bore and bearing fits within 0.0004- and 0.0005-inch tolerances, respectively.

I would bring Foundation hands-on experience spanning mechanical design, simulation, embedded controls, precision fabrication, and hardware validation. I am especially interested in helping develop robotic mechanisms that remain capable and dependable outside the lab, and in learning from a team solving those challenges across the full machine.

Thank you for your consideration.

Sincerely,

Koa Shen